Who will churn, and what that is worth

Logistic Regression + isotonic calibration scored on 1,761 held-out rows that were never used for training or calibration. Probabilities are calibrated, so 0.60 means 60 in 100.

26.5%

BASE CHURN RATE

1,869 of 7,043 customers

0.8495

ROC-AUC, DEPLOYED MODEL

PR-AUC 0.6637; top-decile lift 2.85x

0.1336

BRIER SCORE

was 0.1621 raw, 0.1949 at base rate

981

HIGH-RISK CUSTOMERS

$941,191 of 12-month revenue at risk

Seven models, one split

- Logistic Regression 0.8493 / PR 0.6625 - the deployed model.
- LightGBM tuned with Optuna (15 trials) 0.8472; PyTorch feedforward network 0.8468.
- Everything else: XGBoost 0.837, LightGBM 0.833, Random Forest 0.828, MLP (sklearn feedforward) 0.763.
- The best ensemble, LightGBM (Optuna-tuned) at 0.8472, still lands 0.0021 ROC-AUC below the baseline, so the simplest model is also the best ranker here - and it wins on the two things a client actually needs: calibrated probabilities and coefficients a retention manager can read.
- Calibration is not cosmetic: isotonic scaling moved Brier from 0.1621 to 0.1336 and the deployed artefact is the calibrated pipeline, not the raw model.

Churn is concentrated

- Contract: 42.7% on month-to-month (n=3,875) vs 11.3% one-year and 2.8% two-year.
- Tenure: 47.4% in the first 12 months vs 9.5% beyond 48.
- Payment: 45.3% on electronic check (n=2,365), the largest payment block and the only manual one.
- The 981 customers above the 0.60 cut carry $941,191 of the $5,473,399 12-month book (17%); on the held-out rows, 71.8% of the customers flagged at that cut actually churned.

What saving them is worth (scenario)

- Operating point chosen by expected profit, not accuracy: 0.15 maximises scenario profit on the test split.
- Contacting the top 50% by score returns $74,611 in the scenario.
- Assumptions: 30% save rate, $40 per contacted customer, CLV = 12 x monthly charges.
- These are scenarios, not measured uplift. Neither feed carries an offer, campaign or experiment field, so the save rate is an input, not a result. The first piece of new data to buy is a randomised hold-out.

35/35 data-quality checks passed (Pandera schema + manual) - 144 pytest tests - 78-cell executed notebook with 0 errors - 7-page Streamlit dashboard.

Why they leave, and where to intervene

Survival analysis run in R (survival 3.8.6) and cross-checked against Python lifelines; drivers from SHAP on the tuned LightGBM and permutation importance on the deployed model.

35 months

MEDIAN SURVIVAL, MONTH-TO-MONTH

One-year and two-year: not reached in follow-up

5.55x

HAZARD RATIO, CONTRACT RISK

95% CI 4.89-6.30, p = 4.7e-154

0.60x

SERVICE COUNT IS PROTECTIVE

each additional service bundled

45.0%

OF CHURNERS LEFT TO A COMPETITOR

841 of 1,869, from the 11.1.3 modules

Drivers the client can act on

- Tree SHAP: month-to-month contract 0.636 mean |SHAP|, tenure 0.278, no online security 0.223.
- Permutation importance (PR-AUC drop): log total charges 0.180, internet service 0.088, monthly charges 0.064.
- Cox PH: contract risk 5.55x, bill shock 2.29x, electronic check 1.64x, 1.025x per extra $1 a month.
- R and lifelines agree to 2.4e-06 on every hazard ratio. The proportional-hazards test rejects for contract risk, so read these as average effects over follow-up, not constant ones.

Who the model serves

- Senior citizens are flagged 27.3% of the time vs 10.6% for non-seniors (2.58x), while their observed churn is 42.7% vs 23.4% (1.82x).
- Recall at the cut: 45.9% senior vs 32.5% non-senior (+13.4pp). The segment that churns more receives more of the offers it qualifies for.
- Calibration holds either way: 0.42pp vs 0.07pp mean-prediction gap. Gender: 5.4pp recall spread, under 1pp calibration gap.
- Flagging here means receiving an offer, so unequal selection is a benefit-allocation question, not a denial. It is still checked per segment rather than assumed benign.

Guardrails, and what would break this

- Drift measured on a genuinely different feed (IBM 11.1.3 re-release): score PSI 0.0201 says stable, PaymentMethod PSI 3.39 says RETRAIN.
- The re-release deletes the electronic-check category: 2,365 rows re-map to bank withdrawal and electronic_check_flag drops from 0.336 to 0.000. Realised ROC-AUC still 0.8645 - ranking survived, the feature did not.
- IBM's own Churn Score reaches 0.945 AUC vs our 0.8495, but it encodes the label. It is excluded from every model and used only as an out-of-sample yardstick (Spearman 0.362).
- Next: randomised hold-out to price the offers, a dated feed with a time-based split, and an upstream agreement on the payment and contract re-codes before scoring reaches production.

Neither feed carries support calls, logins, usage, complaints or payment-delay fields. Bill shock, contract-remaining and RFM are proxies built from the 21 columns that do exist, and are labelled as such.